

Security Incident Report

Linux Syslog Security Monitoring & Authentication Log Analysis

Name: Nitin Sukthe

Role: Future SOC Analyst

Environment: Ubuntu Linux Virtual Machine (Lab Environment) with rsyslog enabled and authentication logging configured

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Section 1: Identify the Security Events Involved in the Investigation

The primary security events involved in this investigation were authentication failures, privileged account access, sudo activity, and user session events recorded within Linux authentication logs. These events were identified through analysis of the rsyslog logging service, the system log (syslog), and the authentication log (auth.log).

The investigation focused on authentication-related activities because unauthorized access attempts and privileged account usage are among the most common indicators reviewed by Security Operations Center (SOC) analysts. Authentication failures were observed within auth.log, indicating unsuccessful login attempts. Additional log entries revealed administrative activity through sudo usage and successful root account session creation.

The Linux rsyslog service was verified to be operational and actively collecting security events from the operating system. Authentication activity was then analyzed using Linux command-line tools including grep, awk, less, sort, and uniq to identify security-relevant events and establish an investigation timeline.

The primary goal of this investigation was to determine whether authentication failures represented malicious activity and whether privileged account access occurred following unsuccessful login attempts.

Section 2: Document the Incident

The investigation began after reviewing Linux authentication logs for security-relevant events. During routine log analysis activities, multiple authentication failure events were observed within the auth.log file. These failures indicated that several login attempts were rejected due to incorrect credentials.

The cybersecurity analyst first verified that the rsyslog logging service was operational and actively collecting system logs. The analyst then reviewed the rsyslog configuration file to confirm that authentication and system events were being properly logged. Following configuration validation, the

analyst examined the /var/log directory and identified the primary log sources used throughout the investigation.

The analyst reviewed syslog entries to establish normal operating system activity and then performed targeted searches within auth.log to identify authentication events. Using grep and other filtering techniques, the analyst identified failed login attempts, sudo activity, root account sessions, and user session creation events. Authentication failure entries revealed multiple incorrect password attempts, while sudo and root session entries confirmed successful privileged account access during the testing period.

The analyst then performed event correlation between authentication failures and administrative account activity. Log entries showed three incorrect password attempts followed by successful root account session creation. Further review confirmed that these activities were generated during authorized laboratory testing and were not associated with malicious activity or system compromise.

The investigation determined that the authentication failures were caused by intentional testing performed to generate security events for analysis. No evidence of persistence, privilege escalation abuse, malware execution, account compromise, or unauthorized access was identified. The environment remained secure throughout the investigation.

Section 3: Security Findings

Finding 1: Authentication Failures Detected

The analyst identified multiple authentication failure events within auth.log.

Evidence observed:

- Authentication failure
- Incorrect password attempts
- Failed login activity

These events indicate behavior commonly associated with password guessing and brute-force attacks. Although the events were generated intentionally within the laboratory environment, similar activity would require investigation in a production environment.

MITRE ATT&CK Mapping

Technique ID: T1110

Technique Name: Brute Force

Description:

Adversaries may attempt to gain access by repeatedly guessing passwords until successful authentication occurs.

Finding 2: Privileged Account Access Identified

Root account activity was successfully identified through authentication logs.

Evidence observed:

- Session opened for user root
- Privileged account activity
- Administrative session creation

The presence of privileged account access highlights the importance of monitoring administrative accounts and auditing elevated privileges.

MITRE ATT&CK Mapping

Technique ID: T1078

Technique Name: Valid Accounts

Description:

Adversaries may abuse legitimate credentials to gain and maintain access to systems.

Finding 3: Administrative Activity Recorded

The analyst identified multiple sudo-related events indicating successful administrative actions.

Evidence observed:

- sudo session opened
- sudo session closed

Administrative activity was properly logged and auditable.

MITRE ATT&CK Mapping

Technique ID: T1098

Technique Name: Account Manipulation

Description:

Adversaries may manipulate account permissions or leverage privileged accounts to perform administrative actions.

Finding 4: User Session Monitoring

Authentication logs contained multiple user session creation events.

Evidence observed:

- Session opened
- Session closed

Session monitoring provides visibility into user activity and access patterns across the operating system.

MITRE ATT&CK Mapping

Technique ID: T1033

Technique Name: System Owner/User Discovery

Description:

Monitoring account activity provides visibility into user behavior and account utilization.

Section 4: Risk Assessment

Finding	Risk Level
Authentication Failures	Medium
Root Account Activity	Medium
Administrative Activity	Low
Session Activity	Low

Overall Risk Rating: Medium

The investigation identified authentication failures and privileged account usage, both of which are important indicators of security-relevant activity. While no compromise was identified, these events represent behaviors commonly investigated by SOC analysts in enterprise environments.

Section 5: Recommend Security Improvements

One security improvement recommended by the investigation team is the implementation of Multi-Factor Authentication (MFA). MFA requires users to provide additional verification beyond a password, significantly reducing the risk of unauthorized access caused by stolen or guessed credentials.

Additional recommendations include enforcing strong password policies, configuring account lockout mechanisms after repeated authentication failures, restricting direct root login access, and continuously monitoring privileged account activity.

The organization should also implement centralized log collection and alerting solutions such as Splunk, Wazuh, ELK Stack, Microsoft Sentinel, Google Cloud Logging, or AWS CloudWatch Logs. Automated alerting for authentication failures and privileged account access would improve detection and response capabilities.

These security controls would reduce the likelihood of successful brute-force attacks and strengthen the organization's overall security monitoring posture.

Analyst Conclusion

The Linux Syslog Security Monitoring investigation successfully identified authentication failures, privileged account activity, administrative actions, and user session events through analysis of Linux authentication logs.

The investigation demonstrated practical SOC analyst skills including log analysis, event correlation, threat detection, incident investigation, risk assessment, and MITRE ATT&CK mapping. No indicators of compromise were identified, and all observed activity was consistent with authorized laboratory testing.

The project highlights the importance of authentication monitoring, privileged account auditing, and centralized logging within modern security operations environments.